

curl http://*NODE_IP* :NODE_PORT**/init**

Initialize the database with sample schema

- Now it's time to Insert some sample data. Make sure you will use correct

NODE_IP :*NODE_PORT*

```
#curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "1",  
"user": "John Doe"}' http://NODE_IP :NODE_PORT/users/add
```

curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "2",

"user": "Jane Doe"}' http://*NODE_IP* :*NODE_PORT*/users/add

curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "3",

"user": "Bill Colls"}' http://*NODE_IP* :*NODE_PORT*/users/add

curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "4",

"user": "Mike Taylor"}' http://*NODE_IP* :*NODE_PORT*/users/add

- Now access the data that we have added to database using below command.

curl http://*NODE_IP* :NODE_PORT**/users/1**

- The second time you access the data, it appends '(c)' indicating that it is pulled from the Redis cache.
- Also, try to access mysql shell i.e db pod & run select * from the users table. check app.py for DB related information.
- Prepare proper documentation in brief & write start to end flow. Refer below link if you face any issues.

<https://github.com/janakiramm/Kubernetes-multi-container-pod>

=====

=====

System / Terminal Output

```
payal_kharat@LAPTOP-# git clone https://github.com/janakiramm/Kubernetes-  
multi-container-pod.git  
Cloning into 'Kubernetes-multi-container-pod'...  
remote: Enumerating objects: 51, done.
```

```
remote: Total 51 (delta 0), reused 0 (delta 0), pack-reused 51 (from 1)
Receiving objects: 100% (51/51), 88.14 KiB | 876.00 KiB/s, done.
Resolving deltas: 100% (21/21), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-4$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
```

```
payal_kharat@LAPTOP-# cp -rf Kubernetes-multi-container-pod/* .
payal_kharat@LAPTOP-# ls
Build Deploy Kubernetes-multi-container-pod README.md multi-container-
pod.png
payal_kharat@LAPTOP-# git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Build/
    Deploy/
    Kubernetes-multi-container-pod/
    README.md
    multi-container-pod.png
nothing added to commit but untracked files present (use "git add" to track)
```

```
payal_kharat@LAPTOP-# git add .
warning: adding embedded git repository: Kubernetes-multi-container-pod
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint:   git submodule add <url> Kubernetes-multi-container-pod
```

```
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint:   git rm --cached Kubernetes-multi-container-pod
hint:
hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"
```

```
payal_kharat@LAPTOP-# git commit -m "add Kubernetes multi-container
application"
[master (root-commit) b90b4c4] add Kubernetes multi-container  application
13 files changed, 280 insertions(+)
create mode 100644 Build/Dockerfile
create mode 100644 Build/app.py
create mode 100644 Build/docker-compose.yml
create mode 100644 Build/requirements.txt
create mode 100644 Deploy/db-pod.yml
create mode 100644 Deploy/db-svc.yml
create mode 100644 Deploy/web-pod-1.yml
create mode 100644 Deploy/web-pod-2.yml
create mode 100644 Deploy/web-rc.yml
create mode 100644 Deploy/web-svc.yml
create mode 160000 Kubernetes-multi-container-pod
create mode 100644 README.md
create mode 100644 multi-container-pod.png
```

```
payal_kharat@LAPTOP-# podman build -t cloudethix-multicontainer-python:v1 .
STEP 1/3: FROM python:2.7-onbuild
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:2.7-onbuild...
Getting image source signatures
Copying blob 46dde23c37b3 done      |
Copying blob e7428f935583 done      |
Copying blob 6ebaeb074589 done      |
Copying blob 0c3de61682aa done      |
Copying blob d660b1f15b9b done      |
Copying blob 56f10ddf1173 done      |
Copying blob 4473537c621d done      |
Copying blob 3106f7df3d1c done      |
Copying blob 3de1c6ceef68 done      |
Copying config 3f246dd60a done      |
```

```
payal_kharat@LAPTOP-# podman tag cloudethix-multicontainer-python:v1
payalkharat/cloudethix_multicontainer_payal:python-v1
```

```
payal_kharat@LAPTOP-# podman push
payalkharat/cloudethix_multicontainer_payal:python-v1
Getting image source signatures
Copying blob d660b1f15b9b skipped: already exists
Copying blob 56f10ddf1173 skipped: already exists
Copying blob e7428f935583 skipped: already exists
Copying blob 46dde23c37b3 skipped: already exists
Copying blob 0c3de61682aa skipped: already exists
Copying blob 6ebaeb074589 skipped: already exists
Copying blob 3de1c6ceef68 skipped: already exists
Copying blob 6ba9395573b3 done   |
Copying blob 92c3866eed4c skipped: already exists
Copying blob 8b5ad887043b skipped: already exists
Copying blob 3106f7df3d1c skipped: already exists
Copying blob 4473537c621d skipped: already exists
Copying config 0836e0ab53 done   |
Writing manifest to image destination
```

```
payal_kharat@LAPTOP-# cat web-pod-1.yml
apiVersion: "v1"
kind: Pod
metadata:
  name: web1
  labels:
    name: web
    app: demo
spec:
  containers:
    - name: redis
      image: redis
      ports:
        - containerPort: 6379
          name: redis
          protocol: TCP
    - name: python
      image: payalkharat/cloudethix_multicontainer_payal:python-v1
      env:
        - name: "REDIS_HOST"
          value: "localhost"
      ports:
        - containerPort: 5000
          name: http
          protocol: TCP
```

```
payal_kharat@LAPTOP-# cat web-svc.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: web
  labels:
    name: web
    app: demo
spec:
  selector:
    name: web
  type: NodePort
  ports:
    - port: 80
      name: http
      targetPort: 5000
      protocol: TCP
```

```
payal_kharat@LAPTOP-# kubectl apply -f web-svc.yml
service/web created
```

```
payal_kharat@LAPTOP-# kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	12h
web	NodePort	10.96.196.49	<none>	80:31143/TCP	6s

```
payal_kharat@LAPTOP-# cat db-svc.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: mysql
  labels:
    name: mysql
    app: demo
spec:
  ports:
    - port: 3306
      name: mysql
      targetPort: 3306
  selector:
    name: mysql
    app: demo
```

```
payal_kharat@LAPTOP-# kubectl apply -f db-svc.yml
service/mysql created
```

```
payal_kharat@LAPTOP-# kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
------	------	------------	-------------	---------	-----

kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	12h
mysql	ClusterIP	10.96.91.36	<none>	3306/TCP	5s

payal_kharat@LAPTOP-# cat db-pod.yml

apiVersion: "v1"

kind: Pod

metadata:

name: mysql

labels:

name: mysql

app: demo

spec:

containers:

- name: mysql

image: mysql:5.7.25

ports:

- containerPort: 3306

protocol: TCP

env:

-

name: "MYSQL_ROOT_PASSWORD"

value: "password"

payal_kharat@LAPTOP-# kubectl apply -f db-pod.yml

pod/mysql created

payal_kharat@LAPTOP-# kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
mysql	1/1	Running	0	19s

payal_kharat@LAPTOP-# kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
mysql	1/1	Running	0	88s
web1	2/2	Running	1 (36s ago)	102s

payal_kharat@LAPTOP-# kubectl get svc

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	4m52s
mysql	ClusterIP	10.96.233.135	<none>	3306/TCP	2m34s
web	NodePort	10.96.48.40	<none>	80:32026/TCP	2m48s

payal_kharat@LAPTOP-# curl http://localhost:8080/init

DB Init done

payal_kharat@LAPTOP-# curl -i -H "Content-Type: application/json" -X POST -d
'{"uid":"1","user":"John Doe"}' http://localhost:8080/users/add

HTTP/1.0 200 OK

Content-Type: application/json

Content-Length: 5

Server: Werkzeug/1.0.1 Python/2.7.15

Date: Thu, 03 Sep 2026 09:07:43 GMT

Added

```
payal_kharat@LAPTOP-# curl -i \ $ curl -i \
-H "Content-Type: application/json" \
-X POST \
-d '{"uid":"2","user":"Jane Doe"}' \
http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:11:45 GMT
```

```
payal_kharat@LAPTOP-# curl -i \ -i \
-H "Content-Type: application/json" \
-X POST \
-d '{"uid":"3","user":"Bill Colls"}' \
http://localhost:8080/users/add
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:11:45 GMT
```

```
payal_kharat@LAPTOP-# curl -i \ -i \
-H "Content-Type: application/json" \
-X POST \
-d '{"uid":"4","user":"Mike Taylor"}' \
http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:12:37 GMT
```

```
payal_kharat@LAPTOP-# curl http://localhost:8080/users/1
John Doe
payal_kharat@LAPTOP-# curl http://localhost:8080/users/1
John Doe(c)
payal_kharat@LAPTOP-# curl http://localhost:8080/users/2
Jane Doe
payal_kharat@LAPTOP-# curl http://localhost:8080/users/3
Bill Colls
payal_kharat@LAPTOP-# curl http://localhost:8080/users/4
Mike Taylor
```

```
-----  
payal_kharat@LAPTOP-# kubectl exec -it mysql -- bash  
root@mysql:/# mysql -u root -p  
Enter password:  
Welcome to the MySQL monitor.  Commands end with ; or \g.  
Your MySQL connection id is 4  
Server version: 5.7.25 MySQL Community Server (GPL)
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> SHOW DATABASES;  
+-----+  
| Database          |  
+-----+  
| information_schema |  
| USERDB            |  
| mysql             |  
| performance_schema |  
| sys               |  
+-----+  
5 rows in set (0.01 sec)
```

```
mysql> SHOW DATABASES;  
+-----+  
| Database          |  
+-----+  
| information_schema |  
| USERDB            |  
| mysql             |  
| performance_schema |  
| sys               |  
+-----+  
5 rows in set (0.01 sec)
```

```
-----  
mysql> USE USERDB;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A
```

Database changed

```
mysql> SHOW TABLES;  
+-----+
```



```
| Tables_in_USERDB |
+-----+
| users            |
+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM users;
+-----+-----+
| ID    | USER          |
+-----+-----+
| 1     | John Doe      |
| 2     | Jane Doe      |
| 3     | Bill Colls    |
| 4     | Mike Taylor   |
+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

Question 5

Question

Question-5:

Create 1 Public Docker Hub registry named cloudethix_Initcontainer_yourname.

- Clone below repository on your system.
<https://github.com/janakiramm/simpleapp.git>
- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
- Once code is copied , go to the Build directory and build docker image from docker file and add meaningful tags and push to docker hub repository.
- Once Images are pushed to Docker hub registries, create a directory named kube. Inside the kube directory create deployment.yaml file with 3 replication , label app: simpleapp-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.
- Create one service.yaml file with type nodeport & select simpleapp-webapp pod with port 80 & targetPort 80 with any nodePort between range 30000-32768.
- Open the webpage in the browser and notice the changes and capture the snap.
- Then delete the deployment that you have just created.
- Update the deployment.yaml file and add volumeMounts with mountPath /usr/share/nginx/html from emptyDir: {} volume.

- Once above changes are added, add initContainers block with below parameters. Also add volumeMounts for Init Container with mountPath "/work-dir" from emptyDir: {} volume.
- ```
initContainers:
```
- name: install
  - image: busybox:1.28
  - command:
  - wget
  - "-O"
  - "/work-dir/index.html"
  - <http://info.cern.ch>
- ```
volumeMounts:
```
- name: workdir
 - mountPath: "/work-dir"
- Add volumes with emptyDir: {} in deployment.yaml file.
 - Once the deployment.yaml file is ready, create the deployment & access the page in the browser and notice the changes.

```
=====
=====
```

System / Terminal Output

```
payal_kharat@LAPTOP-# git clone https://github.com/janakiramm/simpleapp.git
Cloning into 'simpleapp'...
remote: Enumerating objects: 47, done.
remote: Total 47 (delta 0), reused 0 (delta 0), pack-reused 47 (from 1)
Receiving objects: 100% (47/47), 8.20 KiB | 335.00 KiB/s, done.
Resolving deltas: 100% (9/9), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-5$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
```

```
hint:  git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
```

```
Initialized empty Git repository in
```

```
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/Que-5/.git/
```

```
payal_kharat@LAPTOP-# cp -rf simpleapp/* .
```

```
payal_kharat@LAPTOP-# ls
```

```
Dockerfile  html  simpleapp  wrapper.sh
```

```
-----
payal_kharat@LAPTOP-# podman build -t cloudethix-initcontainer:simpleapp-v1 .
```

```
STEP 1/4: FROM docker.io/library/nginx
```

```
Trying to pull docker.io/library/nginx:latest...
```

```
Getting image source signatures
```

```
Copying blob 02fc02c4ab8d done |
```

```
Copying blob 6310eb16bf42 done |
```

```
Copying blob c12f394dea35 done |
```

```
Copying blob a44b5c8be616 done |
```

```
Copying blob 956faab5efb3 done |
```

```
Copying blob 07db7bf2649b done |
```

```
Copying blob f340c1b7c1d6 done |
```

```
Copying config 5f52b9aca7 done |
```

```
Writing manifest to image destination
```

```
STEP 2/4: COPY wrapper.sh /
```

```
--> f87d805d1caa
```

```
STEP 3/4: COPY html /usr/share/nginx/html
```

```
--> e9bb2a7f6bce
```

```
STEP 4/4: CMD ["/wrapper.sh"]
```

```
COMMIT cloudethix-initcontainer:simpleapp-v1
```

```
--> 8c761ea81c19
```

```
Successfully tagged localhost/cloudethix-initcontainer:simpleapp-v1
```

```
8c761ea81c19787fc1a5e92ecd523314ce8eedc0617127528bfa465386622712
```

```
-----
payal_kharat@LAPTOP-# podman tag cloudethix-initcontainer:simpleapp-v1
```

```
docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
```

```
payal_kharat@LAPTOP-# podman login docker.io
```

```
Username: payalkharat
```

```
Password:
```

```
Login Succeeded!
```

```
-----
payal_kharat@LAPTOP-# podman push
```

```
docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
```

```
Getting image source signatures
```

```
Copying blob 07db7bf2649b skipped: already exists
```

```
Copying blob 02fc02c4ab8d skipped: already exists
```

```
Copying blob a44b5c8be616 skipped: already exists
Copying blob 956faab5efb3 skipped: already exists
Copying blob 6310eb16bf42 skipped: already exists
Copying blob e4a3debe156f done    |
Copying blob 966b8902a23a done    |
Copying blob c12f394dea35 skipped: already exists
Copying blob f340c1b7c1d6 skipped: already exists
Copying config 8c761ea81c done    |
Writing manifest to image destination
```

```
payal_kharat@LAPTOP-# cat deployment.yaml service.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: simpleapp-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: simpleapp-webapp
  template:
    metadata:
      labels:
        app: simpleapp-webapp
    spec:
      containers:
        - name: simpleapp
          image:
            docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
          ports:
            - containerPort: 80
apiVersion: v1
kind: Service
metadata:
  name: simpleapp-webapp-service
spec:
  type: NodePort
  selector:
    app: simpleapp-webapp
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
```

```
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/simpleapp-webapp created
```

```
payal_kharat@LAPTOP-# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
simpleapp-webapp-6564955fc-frj4r	1/1	Running	0	24s
simpleapp-webapp-6564955fc-g65dk	1/1	Running	0	24s
simpleapp-webapp-6564955fc-w8r9b	1/1	Running	0	24s

```
payal_kharat@LAPTOP-# kubectl apply -f service.yaml
```

```
service/simpleapp-webapp-service created
```

```
payal_kharat@LAPTOP-# kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP
simpleapp-webapp-service	NodePort	10.96.232.18	<none>	80:30080/TCP

```
payal_kharat@LAPTOP-# kubectl delete deployment simpleapp-webapp
deployment.apps "simpleapp-webapp" deleted from default namespace
```

```
payal_kharat@LAPTOP-# kubectl get pods
```

```
No resources found in default namespace.
```

```
payal_kharat@LAPTOP-# cat deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: simpleapp-webapp
```

```
spec:
```

```
  replicas: 3
```

```
  selector:
```

```
    matchLabels:
```

```
      app: simpleapp-webapp
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: simpleapp-webapp
```

```
    spec:
```

```
      initContainers:
```

```
        - name: install
```

```
          image: busybox:1.28
```

```
          command:
```

```
            - wget
```

```
            - "-O"
```

```
            - "/work-dir/index.html"
```

```
            - http://info.cern.ch
```

```
          volumeMounts:
```

```

      - name: workdir
        mountPath: "/work-dir"

containers:
  - name: simpleapp
    image:
docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
    ports:
      - containerPort: 80
    volumeMounts:
      - name: workdir
        mountPath: /usr/share/nginx/html

volumes:
  - name: workdir
    emptyDir: {}

```

```

payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml

```

```

deployment.apps/simpleapp-webapp created

```

```

payal_kharat@LAPTOP-# kubectl get deployment

```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
simpleapp-webapp	3/3	3	3	12s

```

payal_kharat@LAPTOP-# kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
simpleapp-webapp-7748bd6596-2tp92	1/1	Running	0	17s
simpleapp-webapp-7748bd6596-f6dv2	1/1	Running	0	17s
simpleapp-webapp-7748bd6596-zw5hn	1/1	Running	0	17s

```

payal_kharat@LAPTOP-# curl http://localhost:8080/

```

```

<html><head></head><body><header>
<title>http://info.cern.ch</title>
</header>

```

```

<h1>http://info.cern.ch - home of the first website</h1>

```

```

<p>From here you can:</p>

```

```

<ul>

```

```

<li><a href="http://info.cern.ch/hypertext/WWW/TheProject.html">Browse the
first website</a></li>

```

```

<li><a href="http://line-
mode.cern.ch/www/hypertext/WWW/TheProject.html">Browse the first website using
the line-mode browser simulator</a></li>

```

```

<li><a href="http://home.web.cern.ch/topics/birth-web">Learn about the birth
of the web</a></li>

```

```

<li><a href="http://home.web.cern.ch/about">Learn about CERN, the physics
laboratory where the web was born</a></li>

```

```

</ul>

```

```

</body></html>

```

Question 6

Question-6:

Create 1 Public Docker Hub registry named cloudeithx_hpa_yourname.

- Clone below repository on your system.
<https://github.com/vivekamin/kubernetes-hpa-example.git>
- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
- Once code is copied , build a docker image from the docker file and add meaningful tags and push to the docker hub repository.
- Once the image is pushed, go to k8s directory and update deployment.yaml file with image name from your repo. And then create it.
- Open service.yml and change the type to nodePort and apply the same.
- Open the HPA.yaml file, notice it and then apply the same.
- Open the browser, and access the webpage.
- Now it's time to test the HPA working with the below command.

kubectrl run -i --tty load-generator --rm --image=busybox --restart=Never --

```
/bin/sh -c "while sleep 0.01; do wget -q -O- http://NODE_PORT_SERVICE_NAME; done"
```

- Check the HPA from kubectrl command and also check if the deployment is scaling up.
- Take the snap , prepare a well formatted doc and write your understanding.

```
=====
```

System / Terminal Output

```
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
```

```
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/Que-6/.git/

-----

payal_kharat@LAPTOP-# git clone https://github.com/vivekamin/kubernetes-hpa-
example.git
Cloning into 'kubernetes-hpa-example'...
remote: Enumerating objects: 26, done.
remote: Total 26 (delta 0), reused 0 (delta 0), pack-reused 26 (from 1)
Receiving objects: 100% (26/26), done.
Resolving deltas: 100% (9/9), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-6$ cp -rf kubernetes-hpa-example/* .
payal_kharat@LAPTOP-V1NV0QFT:Que-6$ ls
Dockerfile  README.md  k8s        kubernetes-hpa-example  package.json  src

-----

payal_kharat@LAPTOP-# podman build -t cloudehix-hpa:v1 .
STEP 1/6: FROM node:8.12.0-alpine
Resolved "node" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/node:8.12.0-alpine...
Getting image source signatures
Copying blob eeb7d76f44e7 done    |
Copying blob e35f88fcc259 done    |
Copying blob 4fe2ade4980c done    |
Copying config df48b68da0 done    |
Writing manifest to image destination
STEP 2/6: RUN mkdir -p /usr/src/app
--> 5adf40ad5b18
STEP 3/6: WORKDIR /usr/src/app
--> 57b6a4fbd200
STEP 4/6: COPY . /usr/src/app
--> 4fe5d69ed2ef
STEP 5/6: RUN npm install
npm notice created a lockfile as package-lock.json. You should commit this
file.
npm WARN node-app@1.0.0 No repository field.

-----

payal_kharat@LAPTOP-# podman tag cloudehix-hpa:v1
payalkharat/cloudehix_hpa_payal:hpa-v1
payal_kharat@LAPTOP-# podman push payalkharat/cloudehix_hpa_payal:hpa-v1
```



```
Getting image source signatures
Copying blob 93c832ae2e0b done   |
Copying blob 6fb9934f1d99 done   |
Copying blob 94a04f6c7594 done   |
Copying blob 4fe2ade4980c skipped: already exists
Copying blob eeb7d76f44e7 skipped: already exists
Copying blob e35f88fcc259 skipped: already exists
Copying config 2d73be4325 done    |
Writing manifest to image destination
```

```
payal_kharat@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: node-example
  labels:
    app: node-example
spec:
  replicas: 1
  selector:
    matchLabels:
      app: node-example
  template:
    metadata:
      labels:
        app: node-example
    spec:
      containers:
        - name: node-example
          image: payalkharat/cloudethix_hpa_payal:hpa-v1
          imagePullPolicy: Always
          ports:
            - containerPort: 3000
          resources:
            limits:
              cpu: "0.5"
            requests:
              cpu: "0.25"
```

```
payal_kharat@LAPTOP-# kubectl apply -f deployment.yml
deployment.apps/node-example created
payal_kharat@LAPTOP-# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
node-example-65bb46b959-2pk4t	1/1	Running	0	2m41s

```
payal_kharat@LAPTOP-# cat service.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: node-example
  labels:
    app: node-example
spec:
  selector:
    app: node-example
  ports:
    - port: 3000
      protocol: TCP
      nodePort: 30001
  type: NodePort
```

```
payal_kharat@LAPTOP-# kubectl apply -f service.yml
```

```
service/node-example created
payal_kharat@LAPTOP-V1NV0QFT:k8s$ kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	118m
node-example	NodePort	10.96.107.252	<none>	3000:30001/TCP	27s

```
payal_kharat@LAPTOP-# cat hpa.yml
```

```
apiVersion: autoscaling/v1
kind: HorizontalPodAutoscaler
metadata:
  annotations:
  name: node-example
  namespace: default
spec:
  maxReplicas: 4
  minReplicas: 1
  scaleTargetRef:
    apiVersion: extensions/v1
    kind: Deployment
    name: node-example
  targetCPUUtilizationPercentage: 1
```

```
payal_kharat@LAPTOP-# kubectl apply -f hpa.yml
```

```
horizontalpodautoscaler.autoscaling/node-example created
```

```
payal_kharat@LAPTOP-# kubectl get hpa
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS
node-example	Deployment/node-example	cpu: <unknown>/1%	1	4
REPLICAS	AGE			
0	13s			

```
payal_kharat@LAPTOP-V1NV0QFT:Deploy$ curl http://localhost:8080
Hello World!!
```

```
-----
payal_kharat@LAPTOP-# kubectl run -i --tty load-generator --rm --image=busybox
--restart=Never -- /bin/sh -c "while sleep 0.01; do wget -q -O- http://node-
example:80; done"
```

All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command prompt.

```
-----
payal_kharat@LAPTOP-# kubectl get hpa -w
NAME          REFERENCE          TARGETS          MINPODS  MAXPODS
REPLICAS  AGE
node-example  Deployment/node-example  cpu: <unknown>/1%  1         4
0          19m
```

Question 7

Question-7:

Create 1 Public Docker Hub registry named cloudethix_cronjob_yourname.

- Initialize a local repository & copy below code (three files) to your local repository in any of your working branch.
- Once code is copied, build the docker image from Dockerfile , add meaningful tags and then push the docker image to Docker hub registry.
- Now update the pythoncronjob.yml file to change the image name that you have just pushed to docker hub registry.
- Now create a cron job using pythoncronjob.yml file. Check with kubectl command if the cron job is created.
- Check the Job name which is created by cronjob from command line or lens.
- Then check the pod logs which are created by the job and capture the output.
- Prepare well formatted documents and write your understanding.

```
=====
```

System / Terminal Output

```
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
```

```
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/Que-7/.git/
```

```
payal_kharat@LAPTOP-# podman build -t cloudethix-cronjob:1.0 .
STEP 1/7: FROM python:3.7-alpine
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:3.7-alpine...
Getting image source signatures
Copying blob 4819c95424fc done    |
Copying blob ea1518237b37 done    |
Copying blob 9875af95546d done    |
Copying blob 148762f75a1f done    |
Copying blob 96526aa774ef done    |
Copying config 1bac8ae77e done    |
Writing manifest to image destination
STEP 2/7: RUN addgroup -S appgroup && adduser -S appuser -G appgroup
--> 5951b5d6bc02
STEP 3/7: WORKDIR /app
--> bf7dfd535b22
STEP 4/7: COPY helloworld.py /app/helloworld.py
--> 7610f87d17b6
STEP 5/7: RUN chmod +x /app/helloworld.py
--> 467262f4ebfb
STEP 6/7: USER appuser
--> 8c5887a5bfe7
STEP 7/7: ENTRYPOINT ["python", "/app/helloworld.py"]
COMMIT cloudethix-cronjob:1.0
--> 4a23be180315
Successfully tagged localhost/cloudethix-cronjob:1.0
4a23be180315a733cd4b9363c761a666a3cce839df65009f864929e17585964e
```

```
payal_kharat@LAPTOP-# podman tag cloudethix-cronjob:1.0
payalkharat/cloudethix_cronjob_payal:cronjob-v1
payal_kharat@LAPTOP-# podman push
```

```
payalkharat/cloudethix_cronjob_payal:cronjob-v1
Getting image source signatures
Copying blob 8f516151e934 done    |
Copying blob 9875af95546d skipped: already exists
Copying blob ea1518237b37 skipped: already exists
Copying blob 96526aa774ef skipped: already exists
Copying blob 4819c95424fc skipped: already exists
Copying blob 148762f75a1f skipped: already exists
Copying blob 4ab828addb74 done    |
Copying blob 4ab828addb74 done    |
Copying config 4a23be1803 done    |
Writing manifest to image destination
```

```
payal_kharat@LAPTOP-# cat pythoncronjob.yml
```

```
apiVersion: batch/v1
kind: CronJob
metadata:
  name: python-helloworld
spec:
  schedule: "*/1 * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: python-helloworld
              image: payalkharat/cloudethix_cronjob_payal:cronjob-v2
              command: ["/app/helloworld.py"]
          restartPolicy: OnFailure
```

```
payal_kharat@LAPTOP-# kubectl apply -f pythoncronjob.yml
```

```
cronjob.batch/python-helloworld created
```

```
payal_kharat@LAPTOP-# kubectl get cronjob
```

NAME	SCHEDULE	TIMEZONE	SUSPEND	ACTIVE	LAST SCHEDULE
python-helloworld	*/1 * * * *	<none>	False	0	27s

```
payal_kharat@LAPTOP-# kubectl get jobs
```

NAME	STATUS	COMPLETIONS	DURATION	AGE
python-helloworld-29807272	Complete	1/1	9s	2m52s
python-helloworld-29807273	Complete	1/1	3s	112s
python-helloworld-29807274	Complete	1/1	3s	52s

```
payal_kharat@LAPTOP-# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

python-helloworld-29807272-cmp74	0/1	Completed	0	2m57s
python-helloworld-29807273-zbknn	0/1	Completed	0	117s
python-helloworld-29807274-cgsgq	0/1	Completed	0	57s